Class 1

Quantum Field Theory I

Canonical Formalism through Tree-Level QED

Duration: 3 months · 13 weeks · ~5–10 hrs/week
Part of the 2026–2027 Self-Study Plan · RHI → HEP Transition

Builds the canonical apparatus of QFT from classical fields to tree-level QED calculations. Pair with any of the other five classes.

Course Overview

This is the first of two QFT classes. Over 13 weeks you will build the canonical formulation of quantum field theory from classical field theory up through tree-level QED cross sections. By the end you should be able to write down the Feynman rules for QED from a Lagrangian and calculate every canonical $2 \to 2$ tree-level process in your sleep.

The pedagogical line is Peskin & Schroeder Chapters 1–6, with Schwartz and Tong's notes used wherever P&S feels cryptic. Coming from relativistic heavy-ion physics, you already have intuition for QCD and collider phenomenology; this class fills in the formal scaffolding that typical RHI self-training skips. That scaffolding is what makes the HEP theory literature readable.

The second QFT class picks up with path integrals, renormalization, non-Abelian gauge theory, and the electroweak sector. Treat the two together as a one-year QFT sequence compressed into six months of serious part-time work.

Learning Goals

- Quantize free scalar, Dirac, and electromagnetic fields canonically.
- Derive Feynman rules for QED from the Lagrangian, not from memory.
- Compute cross sections for QED $2 \to 2$ processes: $e^- \mu^-$, $e^+ e^- \to \mu^+ \mu^-$, Compton, Bhabha, and Møller scattering.
- Use LSZ reduction and spinor-helicity techniques with fluency.
- Read current HEP theory papers at the level of their kinematic and tree-level-amplitude discussions.

Prerequisites & Background Check

As an active RHI physicist you probably have most of these; the ones to audit are group theory and classical field theory.

- Graduate-level quantum mechanics (Sakurai or Shankar level).
- Classical mechanics in Lagrangian and Hamiltonian form.
- Special relativity as a working tool: four-vector manipulation, Lorentz transformations.
- Basic group theory: SU(2) and its representations, idea of Lie algebras (refresh with Georgi Chs. 1–5 if rusty).
- Complex analysis: contour integration, residues.

Textbooks & Core Resources

Primary Textbooks

- **Peskin & Schroeder**, *An Introduction to Quantum Field Theory*, Chapters 1–6. The canonical spine of this class.
- **Schwartz**, *Quantum Field Theory and the Standard Model*, Parts I–II. Cleaner exposition of many topics; pair with P&S chapter by chapter.

Supplementary / Reference

- **Srednicki**, *Quantum Field Theory* — free draft online; particularly good for spin-0 first approach.
- **Tong**, *Lectures on Quantum Field Theory* (Cambridge, online) — single best free introduction.
- **Weinberg**, Vol. I — reference for axiomatic structure and the origin of the Dirac equation.
- **Zee**, *QFT in a Nutshell* — heuristic and fast; read Parts I–II for physical intuition.
- **Georgi**, *Lie Algebras in Particle Physics* — group theory companion.

Online Lectures & Notes

- David Tong's QFT lecture notes & video lectures (Cambridge).
- Timo Weigand QFT lecture notes (Heidelberg).
- Mark Srednicki's free PDF draft.
- Flip Tanedo's QFT teaching page (UCR) — excellent supplementary problems and explanations.
- CERN Academic Training introductory QFT lectures (recorded, on CDS).

Key Papers & Reviews

- Dirac, *The Quantum Theory of the Electron* (1928) — read for historical intuition once spinors have been seen.
- Feynman, *Space-Time Approach to Quantum Electrodynamics* (1949) — read after Week 8.
- PDG mini-review on “Kinematics” — reference throughout.

13-Week Schedule

Two P&S chapters per ~3 weeks is the rhythm; weeks 12–13 consolidate rather than push into Ch. 7.

Week	Topic	Readings / Activities	Deliverable
1	Lorentz group & classical field theory	P&S Ch. 2.1-2.2; Tong Ch. 1; review four-vectors and tensors	Derive KG equation from action
2	Scalar field quantization	P&S Ch. 2.3-2.4; Schwartz Ch. 2-3	Mode expansion; compute [φ(x), &ph
3	Propagators & causality	P&S Ch. 2.4; Tong Ch. 2.7-2.9	Feynman propagator from contour integr
4	Lorentz group representations; Weyl & Dirac spinors	P&S Ch. 3.1-3.2; Schwartz Ch. 10-11	Show Dirac eqn is Lorentz covariant
5	Dirac equation solutions; discrete symmetries	P&S Ch. 3.3-3.6	C, P, T on Dirac bilinears
6	Dirac field quantization	P&S Ch. 3.5; Schwartz Ch. 12	Quantize with anticommutators; spin-stat
7	Interacting fields & Dyson series	P&S Ch. 4.1-4.3	Derive interaction-picture S-matrix
8	Wick's theorem & Feynman diagrams	Reading: P&S Ch. 4.3-4.4; Schwartz Ch. 7	Write out φ<super>4</super> diagra
9	QED Feynman rules	P&S Ch. 4.8 and Ch. 5.1	Derive QED rules from L_{QED}
10	e<super>+</super>e<super>-</super> & Compton scattering	P&S Ch. 5.1; trace technology (App. A)	Compute e<super>+</super>e<super>-</super> → μ⁺μ⁻ unpolarized cross section fully
11	Compton scattering & crossing symmetry	P&S Ch. 5.5; Schwartz Ch. 13	Klein-Nishina formula
12	Bhabha & Møller; LSZ reduction formula	P&S Ch. 4.6; Schwartz Ch. 6	LSZ derivation written out
13	Review & capstone	Re-read all derivations; work selected problems	Capstone writeup

Pacing assumes ~5–10 hrs/week on this course while running in parallel with one other class. If time is tighter, cut problem sets first; if time is looser, add problems and deeper paper reading.

Problem Sets & Exercises

Problem-solving is where canonical QFT actually sticks. Pick 2–3 problems per P&S chapter; do not skip Chapters 4 and 5.

- P&S Problem 2.1 (classical EM as field theory).
- P&S Problem 2.2 (complex scalar field and charge current).
- P&S Problem 3.1 (Lorentz invariance of Dirac Lagrangian).
- P&S Problem 3.4 (Majorana fermions — foreshadows neutrino physics).
- P&S Problem 4.1 (creation of a classical field by a source).
- P&S Problem 4.3 (linear sigma model — spontaneous breaking warmup).
- P&S Problem 5.1 (Coulomb scattering).
- P&S Problem 5.3 (positronium lifetimes).
- Schwartz's problems at ends of Chs. 10–13 complement P&S well on spinors and tree-level amplitudes.

Capstone Project

Write a ~6–8 page LaTeX document that derives $e^+e^- \to q\bar{q}$ at tree level from the QED Lagrangian, all the way to a numerical cross section as a function of $\sqrt{s}$. Treat the quark as a charged Dirac fermion; color factor appears as N_c.

- Write L_{QED} explicitly for electron and quark; identify all fields and coupling.
- Extract Feynman rules; draw diagrams.
- Compute the amplitude using trace technology; sum over spins.
- Derive $d\sigma/d\Omega$ and integrate; include the factor of N_c from color.
- Plot $\sigma(\sqrt{s})$ in Python for $\sqrt{s}$ from $2m_q$ to 200 GeV for u, d, c, b quarks; compare to LEP R-ratio data.
- Discuss where the calculation breaks down (threshold, QCD corrections) — this sets up QFT II and the Jets class.

Milestones & Self-Assessment

At each checkpoint, you should be able to do the listed tasks without looking them up. If you can't, go back rather than pushing forward.

- **End of Week 4:** You can quantize the free real scalar field from the action alone, and explain why $[\phi(x), \phi(y)]=0$ outside the light cone.
- **End of Week 6:** You can write down the Dirac Lagrangian, derive the canonical anticommutation relations, and explain why fermions require them.

- **End of Week 9:** You can derive QED Feynman rules from L_{QED} on a blank page without looking at P&S.
- **End of Week 12:** You can compute the unpolarized $e^+e^- \to \mu^+\mu^-$ cross section end to end, including trace identities, without notes.
- **End of Week 13:** You can read a modern HEP theory paper up to the point where loop corrections or non-Abelian physics appears.

End of class plan. Iterate after the first two weeks — what was over-ambitious gets cut; what felt thin gets deepened.